PennState Abington

LION Code Challenge – Tech Week 2026

Official Challenge Brief

The challenge: CampusFix

You have 24 hours to build a working prototype of **CampusFix**, a campus maintenance reporting app. Every team solves the same problem; how you solve it is up to you.

The problem. Broken lights, leaking taps, jammed doors and dead outlets often go unreported on campus. When they are reported, the same issue may be reported many times, and students rarely learn whether anything was done.

Your task. Design and build an app that lets the campus community report facility issues, avoids duplicate reports, and shows everyone the status of each issue until it is fixed.

Who it serves. Students, faculty and staff who report issues, and the facilities staff who receive and resolve them. Your app should work well for both.

Timeline

The 24-hour build window opens when this brief is released and closes at the submission deadline. Late submissions are not judged.

When	What
Wed, Oct 28, 10:00 AM	Challenge released; build window opens
Thu, Oct 29, 10:00 AM	**Submission deadline** (24 hours later)
Thu, Oct 29, afternoon	Organizers review videos and repositories
Fri, Oct 30, 10:30 AM	Project report PDF and printed copy due
Fri, Oct 30, 10:30 AM	Teams arrive and set up at the exhibition
Fri, Oct 30, 11:00 AM – 12:00 PM	Exhibition and judging, Banquet Room, Lares Building

All times are Eastern Time.

What to build

Your prototype must complete one full user journey: a user reports an issue, and it moves through to Resolved. Finish the core features before attempting stretch goals.

Core features (required)

1. **Report an issue** with a title, description, category (for example electrical, plumbing, cleaning, furniture, safety, IT) and a campus location.
2. **Attach a photo** to a report.
3. **Browse and search** reports by category, location or status.
4. **Track status** through at least three stages: Reported, In Progress, Resolved.
5. **Basic duplicate handling:** before submitting, show the user similar existing reports and let them confirm or "upvote" an existing one instead of creating a new one.

Stretch goals (optional, considered under Innovation and Technical Quality)

- Smarter duplicate detection using text similarity, location matching or image comparison
- A facilities staff view to assign, prioritize and update reports
- Notifications when a report you filed or upvoted changes status
- A campus map showing reported issues
- A dashboard of trends, such as the most-reported buildings or average time to resolve
- AI features that add genuine value, such as suggesting a category from the photo or description

Use sample or made-up data. Do not connect to real Penn State systems or collect real personal information.

What to submit

Each team submits 3 items by Thursday, October 29, 10:00 AM. One submission per team.

1. **Demo video (maximum 3 minutes).** Videos over 3 minutes will be judged on the first 3 minutes only. Suggested structure:
    a. 0:00–0:30 — the problem and who your app helps
    b. 0:30–2:30 — live walkthrough of the full journey: report → duplicate check → status updates → resolved
    c. 2:30–3:00 — what makes your solution stand out, its limitations, and next steps
2. **Source code repository** (GitHub or similar), shared publicly or with the organizers.
3. **README** in the repository covering: team members, how to run the app, features built, tools and technologies used, and a link to the live app if you have one.

Video tips: record your screen with narration (Zoom, OBS, Loom or your phone all work). Upload to YouTube as **Unlisted**, or to Google Drive or OneDrive with link sharing set to "anyone with the link can view." Test the link in a private browser window before submitting.

Project report (due Friday, October 30, 10:30 AM)

Each team must prepare a 2–4 page project report using the attached template (Project_Report). You may complete the report after the 24-hour coding deadline, but it must describe only the work completed during the build window. Bring a printed copy to the exhibition on Friday and upload the report as a PDF to the Project Report Submission Form linked below by 10:30 AM on October 30.

Project report submission form: LION Code Challenge 2026 – Project Report Submission – Fill out form

How to Submit Your Project

Submit your demo video, source code repository, and README through the official Project Submission Form linked below by Thursday, October 29, at 10:00 AM Eastern. The form closes automatically at the deadline.

Project submission form: LION Code Challenge 2026 – Project Submission – Fill out form.

The form asks for:

- Team name and the email of the team lead
- Link to your demo video
- Link to your code repository
- Link to your live app, if you have one (optional)
- Confirmation that the work was built during the 24-hour window

Submit working links to your demo video and source code repository. Please verify that the organizers can access all links before submitting. Only one submission per team is required. If needed, the team lead may edit the submission before the deadline.

On-campus exhibition: Friday, October 30

All teams present their apps at the Tech Week exhibition, **11:00 AM – 12:00 PM, Banquet Room, Lares Building**. Please arrive by 10:30 AM to set up.

- At least one team member must attend in person.
- Judges and visitors will come to your table. Prepare a **2-minute pitch** and be ready to answer questions about your design and code.
- A poster or slide is welcome but not required.

Top 3 teams will receive awards during Tech Week. All participants receive a signed certificate.

Rules

- Any platform is allowed: web, mobile, desktop, or no-code/low-code tools.
- **AI tools are allowed.** Disclose any AI tools used and explain how you used and verified their outputs in the project report. You must understand and be able to explain all code you submit.
- Use only sample or made-up data. Do not collect real names, emails, ID numbers or photos of people.
- Your app should be usable by everyone: readable text, good colour contrast, and labelled buttons and images.
- Plagiarism or submitting another team's work leads to disqualification.

Judging

Entries are scored out of 100. Judges review the video and repository before the exhibition, then score the live presentation on Friday.

Criterion	Points	What judges look for
Functionality	25	The full journey works end to end; core features are complete
Innovation	20	Original ideas, especially in duplicate detection and stretch features
Usability and design	15	Clear, easy, accessible interface for reporters and staff
Technical quality	15	Sound code, data model and architecture; clear README and diagrams
Responsible design	10	Thoughtful AI use, privacy and accessibility; honest about limitations
Demo video	5	Clear, within 3 minutes, shows the full journey
Live presentation	10	Clear pitch and confident answers to judges' questions

Ties are broken by the Functionality score.

Questions

Contact the organizers, Dr. Iqra Ameer (iqa5148@psu.edu) and Dr. Gina Mejía-Madrid (gzm5448@psu.edu). We will answer questions during the build window.

Can we build on the Lost-and-Found sample if we prepared one? You may reuse general skills and boilerplate, but all CampusFix features must be built during the 24 hours.

Does the app have to be deployed online? No. Running locally is fine, as long as you can demo it live on Friday and in your video.

What if we don't finish everything? Submit what works. A complete core journey scores higher than many half-finished features.

Good luck, and have fun building!